Parameter Golf

Complete Experiment Log

Ciprian-Florin Ifrim
March 2026

This document records 250+ experiments conducted for the OpenAI Parameter Golf Challenge: training the best language model that fits in a 16MB artifact under 10 minutes on 8x H100 GPUs. The final ternary submission achieves **1.1570 bpb** sliding (mean over 3 seeds, std=0.0007) using a 10-layer, 768-dimensional U-Net Transformer with BitNet b1.58 ternary quantisation, relu^2 activation, 4x MLP width, and FP8 QAT storage. A binary variant trained for 50k steps reaches **1.1239 bpb** sliding in a 15.67MB artifact. Both submissions fit within the 16,000,000 byte budget.

Challenge Overview

Train the best language model that fits in a 16MB artifact and trains in under 10 minutes on 8×H100 SXM GPUs, evaluated by tokenizer-agnostic bits-per-byte (BPB) compression on the FineWeb validation set.

- **Baseline:** 1.2244 bpb (9L 512d int8+zlib, 1k vocab)
- **External SOTA:** 1.1194 bpb
- **Our best (ternary, valid):** 1.1565 bpb sliding (P2, 10L 768d relu† 4×MLP fp8, EMBED_DIM=254, seed=42, 16.00MB)
- **Our best (binary, unconstrained):** 1.1239 bpb sliding (15L 768d binary relu† 4×MLP fp8, 50k steps / ~2h compute, 15.67MB)
- **Our best (quality, over budget):** 1.1771 bpb (F59, 12L 768d swiglu 3×MLP, 21.96MB)
- **Challenge period:** March 18 – April 30, 2026
- **Compute sponsor:** OpenAI ($1M in compute credits)

The challenge is framed as L(N) optimisation — minimising loss given fixed parameter count N, unconstrained by data, compute, steps, or architecture. Related challenges include NanoGPT Speedrun (L(T): lowest loss given constrained time) and NanoGPT Slowrun (L(D): lowest loss given constrained dataset).

Run Numbering Convention

Prefix	Description
Plain (1, 100)	Dev runs on RTX 5090, 100 steps
R prefix (R1…)	Record runs — 600s on 8×H100, leaderboard-targeted
S prefix (S1…)	Scaling runs — 1500 steps or 300s on 8×H100, controlled sweeps
SB prefix (SB1…)	Binary scaling runs
F prefix (F1…)	Final runs — 600s on 8×H100, official submissions
P prefix (P1…)	Pushed/submission runs — final config pushed to GitHub

Additionally, 20 early architecture iterations were performed on MLX (Mac Studio M1 Ultra, 32GB unified memory) and 2 on MPS (MacBook Pro M1 Pro, 32GB unified memory) for rapid prototyping before GPU scaling.

> ***Note:*** *This document covers ~85 named runs (F, S, R series). An additional ~165 dev runs (plain numbered 1–100, repeated sweeps, smoke tests) were conducted but are not individually listed. Key findings from those runs are incorporated into the sweep tables and decision rationale. Separate synthetic-data notebooks were used to isolate the behaviour of specific techniques (Tversky similarity, linear alternatives, grouped projections) before committing H100 compute.*

Hardware

System	Spec	Notes
Dev	RTX 5090 32GB, single GPU	Triton smem ceiling 101KB/SM; blocks value embeddings and some kernels
Mac (MLX)	Mac Studio M1 Ultra 32GB	MLX early iteration, 20 runs
Mac (MPS)	MacBook Pro M1 Pro 32GB	MPS early iteration, 2 runs
Final	8×H100 SXM 80GB	Primary training platform

Step times at 768d (12L): relu² 2x: 89ms | relu² 3x: 99ms | relu² 4x: 91ms | swiglu 3x: 127ms | leaky relu 3x: 103ms

Step times at 512d: 26L baseline: 149ms → 136ms with FA3 → 127ms with FA3 + fusions + EMBED=256 at 25L

FlashAttention-3 reduced step time by ~9% (~380 free training steps per 600s run).

Kernel fusion optimisations (fused QKV + fused SwiGLU + dataloader + softcap) saved a further ~7-10ms/step.

Width vs depth discovery: 12L 768d at 106ms/step gets ~5640 steps in 600s vs ~4720 steps for 25L 512d — 920 extra steps from the faster per-step time of wider/shallower models. Final 10L 768d 4×MLP at 91.8ms/step gets ~6530 steps.

Architecture: Ternary U-Net Transformer

Quantisation Scheme

BitNet b1.58 ternary quantisation — weights constrained to {–1, 0, +1} with per-group absmean scaling. Approximately 1.6 bits per parameter.

Compression pipeline: Base-3 packing (5 trits/byte) or bitmask packing → LZMA (preset=9). Best method auto-selected per run. Bitmask wins when zero fraction is high.

Quantisation shrinkage fix: When ternary Q contains zeros, `mean(|Q|) < 1.0`, causing scale mismatch on reload. Fix: inflate by `1/mean(|Q|)` during dequantisation. Eliminates all roundtrip gaps.

U-Net Skip Connections

The model uses a U-Net style encoder/decoder structure with learned skip connections. The first `num_layers // 2` blocks (encoder) store their outputs; the second half (decoder) receives these via `x = x + skip_weight[i] * skips.pop()`. This allows the decoder to simultaneously access high-level semantic representations (from deep processing) and low-level token-level features (from early processing), without requiring the decoder to reconstruct low-level information from the compressed residual stream.

Additionally, each block receives `x0` (the original input embedding) via a learned residual mix: `x = mix[0] x + mix[1] x0`, giving every layer direct access to the raw token representation regardless of accumulated residual drift.

For odd layer counts, the decoder receives the larger half (e.g. 27L → 13 encoder + 14 decoder), which is the standard U-Net convention — more processing power applied after skip injection.

Factored Embedding

With `EMBED_DIM=254`, token embedding is `[8192, 254]` instead of `[8192, 768]`, with learned projections `embed_proj` (254→768) and `embed_proj_rev` (768→254) for the tied output head.

EMBED_DIM history: Started at 128 (dev runs), upgraded to 256 after an optimizer coverage fix revealed that the projection layers had not been receiving gradients (–0.024 bpb improvement vs 128 once trained), then trimmed to 254 to fit artifact+code under the 16,000,000 byte budget (~0.0004 bpb cost, 0.00018/dim from 128→256 scaling data).

Fused Operations

Fused QKV: Single `TernaryLinear(dim, dim + 2*kv_dim)`. **Fused SwiGLU/relu²:** Gate and up projections combined into single wide matrix. Combined saving: ~4-6ms/step.

Z-Loss Regularisation

`1e-4 * logsumexp(logits)²` (from PaLM/Gemma) anchors logits near zero, keeping gradients sharp through the ternary STE.

Compression Scheme

Base-3 + LZMA (Primary)

5 trits per byte (1.585 bits/trit), lossless. LZMA at preset=9 achieves ~39% reduction over int8+zlib. Ternary distribution at convergence: ~20–29% zeros, ~35–40% each ±1. The skewed distribution (more zeros) is exploited by LZMA's entropy coding.

Bitmask Compression (Alternative)

Encodes "is this weight zero?" and "if nonzero, is it +1?" as separate bitmasks. Both methods are tried and the smaller is selected automatically. In practice, bitmask and base-3+LZMA produce nearly identical artifact sizes — bitmask wins marginally in some runs (e.g. S72: 15.84MB vs 15.87MB). Zero fraction would need to drop below ~5% for bitmask to provide a clear advantage; our zero fraction ranges from 17–29% at convergence, making bitmask non-competitive.

3D Tensor Support

Conv1d weights (`[dim, dim, kernel]`) are reshaped to 2D before ternary quantisation and restored to original shape on load.

FP8 QAT

Non-ternary parameters (embeddings, projections) stored at fp8 (e4m3) with Quantisation-Aware Training via STE. Halves fp_params storage (~5MB → ~2.5MB). Typical roundtrip gap: 0.001–0.002 bpb.

Submission Runs (P prefix) — Ternary

Configuration: F88 (10L 768d relu² 4×MLP fp8, WD=0, EMBED_DIM=254, 599s wallclock, TEMP=0.90)

Seed	Steps	val_bpb	RT bpb	Sliding bpb	Train Time	Eval Time	Artifact	Budget
1337	6520	1.1825	1.1839	**1.1568**	**599.1s**	**428.7s**	**15.92MB**	**16.00/16.00 MB**
42	6530	1.1816	1.1837	**1.1565**	**599.7s**	**429.3s**	**15.92MB**	**15.99/16.00 MB**
7	6530	1.1823	1.1850	**1.1578**	**599.6s**	**429.0s**	**15.92MB**	**15.99/16.00 MB**
Mean	**6527**	**1.1821**	**1.1842**	**1.1570**	**599.5s**	**429.0s**	**15.92MB**	
Std	**5**	**0.0005**	**0.0007**	**0.0007**	**0.3s**	**0.3s**	**0.00MB**	

All three seeds fit within the 16,000,000 byte budget. The standard deviation of 0.0007 bpb across seeds confirms high reproducibility. All runs achieve $p < 0.001$ improvement over the 1.2244 bpb baseline.

Batch Size Sensitivity (Ternary, 599s wallclock)

Batch Tokens	Steps	ms/step	val_bpb	Sliding bpb	Tokens Seen	Fits Budget
262,144	10,000	49	1.2413	—	2.6B	No
524,288	**6,530**	**92**	**1.1850**	**1.1578**	**3.4B**	**Yes**
1,048,576	3,480	172	1.1925	1.1659	3.5B	No

524k batch tokens is the optimal operating point. Halving the batch (262k) doubles the step count but degrades quality by 0.056 bpb due to noisier gradients interacting poorly with the ternary STE. Doubling it (1M) sees similar total tokens but fewer gradient updates, costing 0.008 bpb.

Current Best Configuration

Ternary: 10L 768d relu² 4×MLP fp8, WD=0, EMBED_DIM=254

```
NUM_LAYERS=10            MODEL_DIM=768           NUM_HEADS=8
NUM_KV_HEADS=4           MLP_MULT=4              VOCAB_SIZE=8192
ACTIVATION=relu2         LOGIT_SOFTCAP=10        SOFTCAP_TYPE=poly
QK_GAIN_INIT=2.25        ROPE_BASE=5000          ROPE_TYPE=yarn
YARN_MAX_LEN=2048        EMBED_DIM=254           TIE_EMBEDDINGS=1
BITNET_GROUP_SIZE=128    FP_STORAGE=FP8          MUON_WD=0.0
MATRIX_LR=0.04           SCALAR_LR=0.02          TIED_EMBED_LR=0.02
MUON_BACKEND_STEPS=3     MUON_MOMENTUM=0.95      WARMDOWN_FRACTION=0.2
MAX_WALLCLOCK_SECONDS=599
SLIDING_EVAL=1           SLIDING_EVAL_STRIDE=16 TEMP_SCALING=1
TRAIN_BATCH_TOKENS=524288
```

Metric	Value
val_bpb (mean)	1.1821
RT bpb (mean)	1.1842
Sliding bpb (mean)	1.1570
Artifact + code	15,992,753–15,995,705 / 16,000,000 bytes
Steps	6520–6530
ms/step	91.8
zero_frac	0.335–0.336
optimal_T	0.90

Metric	Value
Params	73,685,840

Dev Runs (RTX 5090, 100–500 steps)

Phase 0 — Ternary vs Binary (500 steps, 16L 512d, 1k vocab)

Run	Config	val_bpb	RT bpb	Artifact	ms/step
17	Ternary baseline	1.7110	1.7300	23.95MB	1312
18	Binary {–1,+1}	1.7121	1.7316	23.93MB	1309

Ternary wins by 0.0016 bpb. The zero state provides representational benefit.

Phase 1 — Training Techniques (100 steps, 9L 512d, 1k vocab)

Run	Config	val_bpb	RT bpb	Artifact	Notes
19	Ternary 16L 512d baseline	2.3371	2.3793	7.33MB	
20	+ Untie lm_head at 2/3	2.3569	2.3983	8.13MB	Deferred — needs wallclock fix
21	+ Value embeddings	—	—	—	Blocked: RTX 5090 Triton smem
22	+ Smear module	2.3593	2.3985	7.33MB	Deferred — gate needs many steps
23	Baseline 9L 512d	2.4483	2.4768	4.45MB	Switched from 16L
24	+ Polynomial softcap	2.3981	2.4438	4.45MB	–**0.033 rt**
25	+ Seq length schedule	2.4633	2.5106	4.45MB	Deferred — recompile cost
26	+ NorMuon	2.4018	2.4104	4.40MB	–**0.033 rt**, 5× smaller RT gap
27	+ Grad accum delay	2.6298	2.6571	4.40MB	Deferred — needs 2000+ steps

Vocabulary Sweep (100 steps, 9L 512d)

Run	Vocab	val_bpb	RT bpb	Artifact	Notes
23	1024	2.4483	2.4768	4.45MB	Baseline
28	4096	2.0930	2.0974	6.68MB	–0.32 vs 1k
29	**8192**	**1.9946**	**1.9990**	**9.64MB**	**–0.42 vs 1k — largest single win**

8192 vocab locked. The tokeniser merges ~1.57× more aggressively than 1k, directly reducing BPB. Val token count drops from 63.8M (sp1024) to 40.5M (sp8192) for the same 50k documents.

Activation Sweep (100 steps, 9L 512d, 8k vocab)

Run	Activation	val_bpb	RT bpb	Artifact	ms/step
29	relu2	1.9946	1.9990	9.64MB	838
30	relu	1.9846	1.9879	9.63MB	830
31	**SwiGLU**	**1.9704**	**1.9743**	**10.70MB**	**960**
32	SwiGLU + MTP(2)	1.9627	1.9672	10.69MB	1111

SwiGLU with MTP auxiliary loss gives –0.032 bpb but +16% slower. SwiGLU alone gives –0.025 bpb. MTP deferred.

Embedding Factorization Sweep (100 steps, 9L 512d, 8k vocab)

Run	EMBED_DIM	val_bpb	RT bpb	RT gap	Artifact
33a	0 (=512)	1.9931	1.9962	0.003	9.63MB
33d	**128**	**1.9656**	**1.9656**	**0.000**	**9.12MB**
33c	256	2.0538	2.1339	0.080	6.68MB
33e	64	2.0936	2.0968	0.003	4.49MB
33f	1024	2.0709	2.1845	0.114	15.60MB

128 was optimal at dev scale. After an optimizer fix revealed the projection layers had not been training, 256 became optimal at full convergence — see EMBED_DIM Sweep at full convergence.

Tversky Neural Network Investigation

Based on Doumbouya et al. (2025). Three-term Tversky similarity: `S = theta f(A intersection B) - alpha f(A - B) - beta * f(B - A)` with learned membership functions.

Feature count sweep (FP16 features, ternary prototypes, 100 steps, 9L 512d):

Run	Features	val_bpb	RT bpb	RT gap	Artifact
—	No Tversky	1.9751	1.9751	0.000	5.33MB
38	16	1.9877	2.0186	0.031	5.46MB
39	32	1.9843	2.0133	0.029	5.57MB
40	64	1.9790	2.0097	0.031	5.79MB
41	**128**	**1.9427**	**1.9865**	**0.044**	**6.20MB**
42	256	1.9737	2.0863	0.113	5.63MB
43	512	2.0036	2.0965	0.093	5.90MB
44	128 + shrinkage fix	1.9425	**1.9425**	**0.000**	**6.20MB**

Tversky showed genuine quality benefit (~-0.017 bpb) at dev scale with 128 features and fp16 prototype storage. However, subsequent investigation at full convergence (12L 768d) and with corrected prototype storage showed all Tversky variants within noise of the linear baseline. Additional experiments included full ternary prototypes, shared feature pools across layers, no-features mode, logit-head application, and different membership functions (sigmoid, poly, tanh). A synthetic-data notebook confirmed that Tversky's asymmetric similarity only helps on tasks with genuine directional feature relationships (hypernym/hyponym, cause/effect); next-token prediction on FineWeb web text is not such a task.

At the 768d architecture with relu², Tversky also incurred a 19ms/step overhead because the smaller MLP no longer masked the compute cost.

Conclusion: Tversky is quality-neutral on FineWeb language modelling regardless of configuration. Not a quantisation issue, not an optimizer issue — the task simply does not benefit from asymmetric similarity.

Key Hyperparameter Sweeps (100 steps, 9L 512d, 8k vocab)

QK_GAIN_INIT sweep:

Run	QK_GAIN	val_bpb	Delta
75	1.0	2.0007	+0.0076
73	1.5	1.9931	baseline
81	2.15	1.9913	–0.0018
79	**2.25**	**1.9898**	**–0.0033**
77	2.5	1.9915	–0.0016
80	2.75	1.9975	+0.0044
78	3.0	2.0011	+0.0080

Clear inverted-U response. **QK_GAIN_INIT=2.25 locked.**

LOGIT_SOFTCAP sweep:

Run	SOFTCAP	val_bpb	Delta
74	5	1.9942	–0.0013
73	**10**	**1.9931**	**–0.0024**
72	20	1.9935	–0.0020
71	50	1.9957	+0.0003

LOGIT_SOFTCAP=10 locked.

Softcap type (poly vs tanh):

Run	Type	val_bpb	Notes
S23	poly	1.3680	
S24	tanh	1.3693	
S28/S29	both at EMBED=1024	1.3460–1.3462	Identical at convergence

Zero effect. Polynomial retained as default.

ROPE_BASE sweep:

Run	ROPE_BASE	val_bpb	Notes
70	**5000**	**1.9959**	**Best at short training**
73	10000	1.9931	Close second
69	20000	2.0008	
68	50000	2.0017	

KV Heads:

Run	KV_HEADS	val_bpb	Artifact
58	**4 (GQA)**	**1.9955**	**7.75MB**
66	8 (MHA)	2.0148	8.46MB

MLP_MULT:

Run	MLP_MULT	val_bpb	Artifact
58	**2**	**1.9955**	**7.75MB**
64	3	2.0004	9.09MB

Run	MLP_MULT	val_bpb	Artifact
65	4	1.9992	10.39MB

Storage precision:

Run	Storage	val_bpb	RT bpb	RT gap	Artifact
90	**fp16**	**1.9656**	**1.9656**	**0.000**	**9.06MB**
91	fp8	1.9662	1.9702	0.004	7.83MB
92	fp4	1.9661	1.9955	0.029	7.11MB

TTT-LoRA sweep (100 steps, ROPE=5000):

Run	Rank	LR	TTT bpb	Delta
85	**8**	**0.01**	**1.9368**	**–0.0315**
86	8	0.005	1.9378	–0.0312
87	8	0.02	1.9644	–0.0038
88	**4**	**0.01**	**1.9371**	**–0.0285**
89	16	0.01	OOM	—

TTT confirmed working at dev scale (–0.0315 bpb). Incompatible at convergence — see TTT investigation.

EMBED_DIM sweep at 512d (12L, 100 steps):

Run	EMBED_DIM	Tversky feat	RT bpb	Artifact	bpb/MB efficiency
95	64	128	2.1961	8.40MB	worst
98	96	128	2.0356	8.74MB	
97	128	128	1.9656	9.12MB	best
99	192	128	2.0409	10.07MB	
94	256	128	2.0703	10.93MB	
100	256	256	2.0340	10.09MB	RT gap 0.021
96	512 (off)	128	2.0642	13.50MB	

128 confirmed optimal at dev scale.

Architecture Sizing Table (Ternary, EMBED_DIM=128, standard proj)

Config	Layers	Artifact	Under 16MB?	RT gap	Headroom
fp16	20	14.23MB	Yes	0.0001	1.77MB
fp16	**22**	**15.48MB**	**Yes**	**0.0001**	**0.52MB**
fp16	24	16.74MB	No	—	–0.74MB
fp8 QAT	24	14.63MB	Yes	0.028	1.37MB
fp8 QAT	26	15.77MB	Yes	0.066	0.23MB
fp8 QAT	**27**	**15.42MB**	**Yes**	**0.0025**	**0.58MB**
fp8 QAT	28	15.92MB+code	Marginal	0.0029	~0MB
fp8 QAT	30	16.92MB	No	0.0029	–0.92MB

H100 Record Runs (R prefix)

Hardware: 8×H100 SXM 80GB | **Time limit:** 600 seconds

Run	Config	Steps	val_bpb	RT bpb	Artifact	Notes
R1	22L Tversky fp16	4299	1.2789	1.2792	15.80MB	
R2	26L standard fp16	3973	1.2649	1.2650	15.85MB	Pre-LR tuning best
R3	16L Tversky fp16	5949	1.2900	1.2904	11.95MB	Too shallow
R4	9L Tversky fp16	10112	1.3374	1.3394	7.48MB	Way too shallow
R5	30L fp8	2852	1.2689	1.2815	17.22MB	Over budget
R6	26L fp16, 2× LR	~4003	1.2991	—	~15.85MB	LR overshot
R7	**26L fp16, LR=0.02**	**4008**	**1.2608**	**1.2610**	**15.83MB**	**Best pre-FA3**
R8	26L fp16, LR=0.01	4017	1.2853	1.2855	15.72MB	LR too low
R9	26L BigramHash	4010	1.2804	1.2802	15.81MB	BigramHash negative
R10	26L untie@66%	3706	1.2754	1.2753	23.15MB	Over budget
R11	26L tied, updated code	4009	1.2806	1.2808	15.81MB	Code regression

LR sweep (R-series):

LR	val_bpb	Notes
0.08	1.2991	Overshoots — ternary STE amplifies gradient noise
0.02	**1.2608**	**Optimal**
0.01	1.2853	Too slow

Scaling Runs (S prefix)

Hardware: 8×H100 SXM 80GB | **Steps:** 1500 | **Timer:** disabled (MAX_WALLCLOCK_SECONDS=0) **Base config:** 26L 512d, EMBED_DIM=128, ROPE=5000, QK_GAIN=2.25, SOFTCAP=10, LR=0.02 all, VOCAB=8192, SwiGLU, SEED=1337

Warmdown Sweep

Run	Fraction	val_bpb
S3	10%	1.3467
S1	**20%**	**1.3438**
S2	30%	1.3443
S4	30% repeat	1.3458
S5	40%	1.3501

S2 vs S4 (identical config): 0.0015 bpb spread — confirmed seed variance floor.

Muon Backend Steps

Run	Steps	ms/step	val_bpb
S8	3	144.87	1.3491
S9	4	146.61	1.3448
S1	**5**	**149.19**	**1.3438**
S7	8	164.31	1.3441
S6	10	157.95	1.3456

At full convergence (F6 vs F1): 3 steps matches 5 due to +190 extra training steps. Locked at 3.

Muon Momentum

Run	Momentum	val_bpb	zero_frac	Artifact
S11	0.90	1.3680	0.179	15.39MB
S1	**0.95**	**1.3438**	**0.205**	**15.56MB**
S10	0.99	1.3505	0.259	15.78MB

Higher momentum increases zero_frac, inflating artifact size.

Architecture Experiments

Run	Config	ms/step	val_bpb	Notes
S12	20L 640d (80M params)	160.58	1.6676	17.75MB — over budget
S1	**26L 512d baseline**	**149.19**	**1.3438**	**Reference**
S13	26L, TRAINING_DR=2	281.63	1.3727	~795 effective steps, OOM at DR=3

Eval Depth Recurrence Sweep

Run	EVAL_DR	val_bpb
S15	0/1	1.3685–1.3690
S16	2	1.3688
S17	3	1.3681
S18	4	1.3690
S19	5	1.3683

Total range: 0.0009 bpb — pure noise.

Weight Decay (1500 steps)

Run	MUON_WD	val_bpb	zero_frac	Artifact
S15	**0.00**	**1.3685**	**0.179**	**15.39MB**
S20	0.04	1.3722	0.145	15.12MB

WD hurts at 1500 steps but saves 0.27MB. Reversed at full convergence — see Final Ternary Record Runs.

BigramHash

Run	Config	Steps	val_bpb	Artifact
S21	26L + BigramHash	1500	1.3681	15.45MB
R9	26L + BigramHash	4010	1.2804	15.81MB

At full convergence: 0.020 bpb worse than R7. The 2.1MB fp16 cost of the bigram table displaces ternary layer depth at convergence. **Not viable within budget.**

Tied Embedding / Correction Weight / Untie Investigation

TIE_EMBEDDINGS	UNTIE_AT_FRACTION	LM_HEAD_RANK	Behaviour
0	any	any	Untied from start — unstable, loss = log(8192) = 9.01
1	0.0	0	Always tied — current best
1	0.66	0	Tied → full-rank untie at 66% of wallclock
2	0.0	0	Tied + correction weight residual on tok_emb
2	0.66	0	Tied + correction → full-rank untie at 66%
2	0.66	r	Tied + correction → SVD rank-r untie at 66%

1500-step results:

Run	TIE	UNTIE	RANK	val_bpb	Artifact
S15	1	0.00	0	1.3685	15.39MB
S30	2	0.00	0	1.3678	15.39MB
S36	1	0.66	0	1.3648	22.83MB
S37	**2**	**0.66**	**0**	**1.3642**	**22.84MB**
S38	1	0.66	0	1.3667	22.84MB
S39	0	0.66	0	3.4890	10.88MB

Untie gives +0.005 bpb gain but adds 7.3MB — over budget. **TIE=1, no untie locked.**

LM Head Factorization (SVD-at-Untie)

Run	RANK	val_bpb	Artifact	Delta vs baseline
S37	0 (full)	1.3642	22.84MB	+0.004 — over budget
S43	32	1.4873	17.27MB	–0.119
S41	64	1.4243	17.60MB	–0.056
S42	128	1.3889	18.40MB	–0.020

SVD factorization does not recover within the remaining 34% of training. The model requires full-rank lm_head for 8192-class separability in 512-dimensional space.

Tied Embed LR Sweep

Run	TIED_EMBED_LR	MATRIX_LR	SCALAR_LR	val_bpb
S33	0.01	0.02	0.02	1.3723
S15	**0.02**	**0.02**	**0.02**	**1.3685**
S34	0.03	0.02	0.02	1.3742

Symmetric degradation. **TIED_EMBED_LR=0.02 locked.**

TTT-LoRA Investigation

Test-time training with per-document LoRA adapters. Confirmed working at dev scale (–0.0315 bpb). Incompatible at convergence across 6 diagnostic runs.

Run	Config	val_bpb	TTT bpb	Notes
S22	TTT_LR=0.01	1.3690	1.5065	TTT hurts
S23	No lm_head_lora	1.3690	1.4993	Still hurts
S24	tanh softcap	1.3693	1.4982	No improvement
S25	Q/V loras only	1.3692	1.5193	Worse
S26	EMBED_DIM=1024	1.3473	1.4746	Bottleneck not cause
S27	9L (original depth)	1.4039	1.5189	Still incompatible at 9L

Root cause: Every `TernaryLinear` applies RMSNorm to its input before the weight multiply. The LoRA adapter delta is computed on the pre-normalised representation, but injected into a forward pass where base weights operate on a differently-normalised space. At 100 steps the model is poorly calibrated and LoRA signal dominates. At convergence, the base model's representations are precisely calibrated to this normalised space, and any LoRA delta corrupts rather than adapts. This incompatibility is architectural. **TTT permanently disabled.**

MTP (Multi-Token Prediction)

Run	MTP_HEADS	ms/step	val_bpb	Notes
S47	**0**	**149**	**1.3693**	**Baseline**
S45	2	157	1.3704	+0.0011 worse
S62	2	144	1.3727	+0.0034 worse

Confirmed at both 1500 steps and full convergence (post-fix retest: 0.006 bpb worse at both MTP=1 and MTP=2). A 60M+ parameter, 1.58-bit model does not have the parameter bandwidth for auxiliary future-planning objectives.

Smear Module

Run	SMEAR	val_bpb	ms/step
S48	**0**	**1.3687**	**149**
S49	1	1.3675	182

+22% slower, –0.0012 bpb at 1500 steps. At full 600s wallclock, smear costs ~740 fewer training steps. Not viable within the ternary 10-minute budget but explored further in the binary track.

Sequence Length Schedule

Run	Config	val_bpb	ms/step avg
S48	baseline	1.3687	149
S51	smear + seq@33%	1.3660	~240
S52	smear + seq@33% repeat	1.3640	~221
S58	**smear + seq@33% + YaRN**	**1.3628**	**~221**

Real gain at 1500 steps but severe step penalty at full 600s. **Disabled for final runs.**

Batch Size Schedule

Run	Config	val_bpb
S48	baseline	1.3687
S50	smear + batch	1.3698
S53	smear + seq + batch	1.3667

Noisier gradients interfere with ternary STE convergence. **Not viable.**

YaRN Positional Encoding

Run	Config	val_bpb
S48	RoPE baseline	1.3687
S54	YaRN 4096	1.3705
S55	YaRN 2048	1.3679
S56	YaRN 2048 + seq@33%	1.3672
S57	YaRN 2048 + seq@50% + smear	1.3637
S58	**YaRN 2048 + seq@33% + smear**	**1.3628**

YaRN 4096 hurts (scale=0.25 too aggressive). YaRN 2048 marginally better. **YaRN 2048 retained; seq schedule disabled.**

ROPE_BASE with YaRN: S63 (10000) = 1.3692, **S61 (5000) = 1.3686**. ROPE_BASE=5000 locked.

Sliding Window Evaluation

Run	Stride	Sliding bpb	Eval time
S60	16	1.3452*	>600s
S67	24	1.3146	592s
S61/S66	**32**	**1.3139–1.3452***	**~350s**

*S60/S61 used incorrect momentum=0.90. At full convergence (F1): stride=32 gives 1.2312 sliding bpb in 280s.

Temperature Scaling

Grid search over T in [0.80, 1.20] on 65,536 training tokens. 5-point grid. Optimal T was consistently 1.00 at convergence for the 512d SwiGLU architecture. At the 768d relu² architecture, T=0.90 was consistently optimal (relu² logits slightly underconfident). **TEMP_SCALING=1 in all final runs.**

Group Size Sweep (S73–S76, 2000 steps, 27L)

Run	Group Size	Layers	val_bpb	Artifact	Total
S76	32	27	1.2739	17.64MB	17.73MB
S75	64	27	1.2683	16.22MB	16.31MB
S73	**128**	**27**	**1.2677**	**15.53MB**	**15.62MB**
S74	256	27	1.2699	15.19MB	15.28MB

128 wins on both quality and compression.

Skip Weights Init — Zero vs Ones (S77)

Run	Init	val_bpb	artifact
S73	ones	1.2677	15.62MB
S77	zeros	1.2781	15.62MB

Zero-init is **0.0104 bpb worse**. Decoder needs skip signal from step 0.

FP8/FP4 Storage with QAT

FP8 sweep:

Run	Config	val_bpb	RT bpb	RT gap	Sliding bpb	Artifact
S64	26L fp16	1.3390	1.3390	0.000	1.3150	15.58MB
S65	30L fp8, no QAT	1.3346	1.3394	0.0048	1.3150	16.92MB
S66	30L fp8, QAT	1.3351	1.3380	0.0029	**1.3139**	**16.92MB**
S71	27L fp8, QAT	1.3380	1.3405	0.0025	1.3164	15.42MB
S72	28L fp8, QAT	1.3377	1.3406	0.0029	1.3166	15.92MB

QAT reduces fp8 RT gap from 0.0048 to 0.0029 (40% improvement). However at full convergence (F3), 28L fp8 QAT (1.2353 sliding) loses to 26L fp16 (1.2312 sliding).

FP4 sweep:

Run	Config	val_bpb	RT bpb	RT gap	Sliding bpb	Artifact
S68	30L fp4 QAT	1.3377	1.3643	**0.0266**	**1.3404**	**16.49MB**
S69	26L fp4 Tversky QAT	1.3543	1.3835	**0.0292**	**1.3606**	**15.01MB**
S70	28L fp4 QAT	1.3405	1.3666	**0.0261**	**1.3424**	**15.43MB**

FP4 RT gap of ~0.026–0.029 even with QAT is unrecoverable. **FP4 not viable at any layer count.**

EMBED_DIM Sweep (Full Convergence, 25L)

Config	EMBED_DIM	Steps	val_bpb	sliding_bpb	artifact	Notes
S80	0 (=512)	4500	1.1902	~1.168 est	19.78MB	OOM on sliding eval
F22	**256**	**4720**	**1.2012**	**1.1739 (s16)**	**16.21MB**	**Best 512d result**
F16-era	128	4310	1.2245	—	16.19MB	Pre-fix baseline

EMBED_DIM=256 locked. Budget impact: fp_params ~4.85MB vs ~2.48MB at 128 (+2.37MB).

Final Ternary Record Runs (F prefix)

Hardware: 8×H100 SXM 80GB | **FlashAttention-3 enabled** | **Time limit:** 600 seconds

Run	Config	Steps	val_bpb	RT bpb	Sliding bpb	Eval time	Artifact
F1	**26L fp16, no smear, no seq**	**4362**	**1.2560**	**1.2560**	**1.2312**	**280s**	**15.85MB**
F2	26L fp16, smear + seq@33%	3044	1.2779	1.2778	1.2535	390s	15.85MB

Run	Config	Steps	val_bpb	RT bpb	Sliding bpb	Eval time	Artifact
F3	28L fp8 QAT, no smear, no seq	4019	1.2571	1.2601	1.2353 (s24)	385s	16.14MB
F4	26L fp16, EMA=1	4145	1.2589	2.3307	—	—	14.52MB
F5	26L fp16, EMA fix v1 (smoke)	407	1.5483	2.3642	—	—	14.90MB
F6	26L fp16, MUON_BACKEND_STEPS=3	4552	1.2558	1.2558	1.2311 (s24)	362s	15.81MB
F7	26L fp16, WD=0.04, steps=3	4499	1.2552	1.2551	1.2302 (s24)	362s	15.60MB
F8	28L fp16, WD=0.04, steps=2, LR=0.02	4219	1.2799	1.2801	1.2558 (s16)	577s	15.92MB
F9	28L fp16, WD=0.04, steps=2, LR=0.03	4231	1.2673	1.2676	1.2431 (s16)	577s	16.00MB
F10	28L fp16, WD=0.04, steps=2, LR=0.04	4226	1.2636	1.2636	1.2391 (s16)	578s	16.01MB
F11	28L fp16, WD=0.04, steps=3, LR=0.04	4137	1.2489	1.2488	—	—	16.69MB
F12	28L fp16, WD=0.04, steps=4, LR=0.04	4047	1.2496	1.2500	—	—	16.71MB
F13	28L fp16, WD=0.04, steps=3, LR=0.05	4048	1.2512	1.2510	—	—	16.73MB
F14	28L fp16, WD=0.04, steps=3, LR=0.08	4036	1.2576	1.2574	—	—	16.75MB
F15	27L fp16, AdamW matrix, LR=0.01	4676	1.2943	1.2942	—	—	15.71MB
F16	27L fp16, Muon, LR=0.04, WD=0.04	4310	1.2245	—	—	—	16.19MB
F22	**25L fp16, EMBED=256, steps=3, WD=0.04**	**4720**	**1.2012**	**1.2011**	**1.1739 (s16)**	**493s**	**16.21MB**

Key findings: F22 with EMBED_DIM=256 and corrected optimizer achieves 0.055 bpb improvement over F1 (the best pre-fix config). 28L extensively attempted (F8–F14) but artifact always over budget at competitive LR. AdamW for matrix params (F15) is clearly worse than Muon.

Phase 2 — Post-Optimizer-Fix Experiments (25L 512d EMBED=256)

EMA (Exponential Moving Average)

Run	Config	Steps	val_bpb	RT bpb	Artifact
F4	EMA=1, decay=0.999	4145	1.2589	2.3307	14.52MB
—	Full run with EMA	4144	1.2584	1.3776	14.94MB

EMA is fundamentally incompatible with ternary quantization. EMA averaging in fp32 produces smoother, more zero-centered weights. More latent weights near zero → more round to 0 in ternary → scale factor mismatch → 0.13 bpb RT gap. **Permanently disabled.**

Muon Backend Steps — Full Convergence

Run	Steps	step_avg	val_bpb	sliding_bpb	artifact
F1 (steps=5)	4362	137ms	1.2560	1.2312	15.85MB
F6 (steps=3)	4552	131ms	1.2558	1.2311	15.81MB

6ms/step saving → 190 extra steps → quality equivalent. **MUON_BACKEND_STEPS=3 locked.**

Weight Decay — Full Convergence

Run	WD	Steps	val_bpb	sliding_bpb	zero_frac	artifact
F6	0.00	4552	1.2558	1.2311	0.294	15.81MB
F7	0.04	4499	1.2552	1.2302	0.221	15.60MB

WD=0.04 wins at full convergence on the 26L architecture. However at 10L 4×MLP (Phase 4), WD=0.00 was better — wider MLP needs full weight freedom.

MTP Retest (Post-Fix)

Run	MTP_HEADS	Steps	step_avg	val_bpb	artifact
F22 baseline	0	4720	127ms	1.2012	16.29MB
Run 26	1	4560	131ms	1.2074	16.30MB
Run 27	2	4420	135ms	1.2074	16.29MB

MTP confirmed not viable post-fix. 0.006 bpb worse at both heads. **MTP_HEADS=0 permanently locked.**

Tversky Phase 2 (Post-Fix, 12L 768d, fp16 Prototypes)

Comprehensive retest with corrected optimizer and fp16 prototype storage:

Run	Config	Features	Pools	val_bpb	RT gap
49	No Tversky	—	—	**1.1888**	**0.0002**
50	Attn proj only	128	1	1.1893	0.0000
51	Attn proj only	256	1	1.1894	0.0001
52	Attn proj only	32	1	1.1898	0.0001

Run	Config	Features	Pools	val_bpb	RT gap
53	Attn + head	128	1	1.1892	—
54	Attn + head	128	0 (local)	1.1897	+0.0006

All variants within 0.001–0.002 bpb of baseline — pure noise. Confirmed by synthetic-data analysis that Tversky's asymmetric similarity only helps on tasks with directional feature relationships, which next-token prediction on web text is not.

Phase 3 — Architecture Exploration (Post-Optimizer-Fix)

Width vs Depth

The central Phase 3 finding: wider models with fewer layers beat deeper models.

768d Scaling Curve

Run	Layers	Steps	step_avg	val_bpb	Artifact
34	8	8110	74ms	1.2894	12.94MB
30	12	5640	106ms	1.1893	17.50MB
38	14	4900	122ms	1.1870	19.79MB
33/37	16	4320	139ms	1.1825–37	22.08MB
39	18	3870	155ms	1.1801	24.39MB
36	20	3510	171ms	1.1854	26.67MB

Peak at 18L, then step penalty dominates. 8L collapses (U-Net encoder too shallow). Seed variance: Run 33 vs 37 = 0.0012 bpb.

Cross-Architecture Comparison

Config	Layers	Dim	Steps	val_bpb
F22	25	512	4720	1.2012
Run 30	12	768	5640	1.1893
Run 40	8	1024	5870	1.1858
Run 41	10	896	5400	1.1862
Run 35	20	640	4170	1.1927
Run 42	6	896	8510	1.2157

Width beats depth: 12L 768d (1.1893) beats 25L 512d (1.2012). Minimum viable depth: 768d ~10–12L, 896d ~10L, 1024d ~8L.

FP8 at 768d

Run	Layers	Storage	val_bpb	RT bpb	RT gap
49	12	fp16	1.1888	1.1886	0.0002
42	13	fp8	1.1879	1.1900	0.0021

FP8 RT gap acceptable at 768d. Enables extra layers within budget.

LM_HEAD_RANK Investigation (Post-Fix, 768d)

Run	Config	val_bpb	RT bpb	Total	Notes
Run 49	baseline	1.1888	1.1886	17.50MB	Reference
Run 43	TIE=2, rank=256, fp8	1.2021	1.2028	20.41MB	Artifact bloated
Run 44	TIE=0, rank=512, untie=0.0	1.3196	1.3195	16.92MB	Random head, no learning
Run 45	TIE=2, rank=512, fp16	1.2312	1.2317	26.87MB	Catastrophic artifact blowup

Root cause: the SVD factors U and V require fp16/fp8 precision to maintain approximation quality. At any viable compression level, the two new matrices cost more storage than the original tied embedding saves. **Not viable.**

Phase 4 — Final Architecture Search

Activation Sweep (12L 768d 3×MLP, 600s)

Run	Activation	MLP	ms/step	Steps	val_bpb	Artifact
F55	relu	2×	88.7	6760	1.2284	14.49MB
F56	**relu²**	**2×**	**89.5**	**6700**	**1.2042**	**14.48MB**
F60	leaky relu	3×	102.6	5840	1.2094	17.50MB
F57	**relu²**	**3×**	**101.5**	**5910**	**1.1878**	**17.51MB**
F58	swiglu	3×	127.4	4700	1.1786	22.05MB
F59	**swiglu**	**3×**	**127.3**	**4710**	**1.1771**	**21.96MB**

relu² beats relu by 0.024 bpb at no cost — strictly dominant. relu² locked for budget-constrained path.

MLP Width Sweep (600s)

Run	Activation	MLP	Layers	ms/step	Steps	val_bpb	Artifact
F56	relu²	2×	12	89.5	6700	1.2042	14.48MB
F64	relu²	3×	12	99.4	6030	1.1873	17.50MB
F75	relu²	4×	12	91.6	6550	1.1795	20.54MB
F82	relu²	4×	10	91.6	6550	1.1861	16.04MB

4× MLP at 10L beats 3× at 12L within similar budget.

Layer Count vs MLP Width (fp8, 600s)

Run	Config	Layers	ms/step	Steps	val_bpb	RT bpb	Artifact
F78	relu² 3× fp8	12	99.3	6040	1.1884	1.1898	15.80MB
F77	relu² 3× fp8	13	106.6	5630	1.2065	1.2077	16.96MB
F80	relu² 2× fp8	15	106.9	5610	1.2120	1.2136	15.45MB
F81	relu² 2× fp8	16	113.9	5270	1.1996	1.2009	16.33MB
F79	relu² 3× fp8	11	91.5	6560	1.1920	1.1933	14.66MB
F82	**relu² 4× fp8**	**10**	**91.6**	**6550**	**1.1861**	**1.1877**	**16.04MB**
F83	swiglu 3× fp8	10	105.5	5690	1.1842	1.1853	17.29MB

Weight Decay at 10L 4×MLP fp8

Run	WD	val_bpb	RT bpb	Artifact
F82	0.04	1.1861	1.1877	16.04MB
F84	0.08	1.1983	1.1998	16.04MB
F85	**0.00**	**1.1828**	**1.1844**	**16.02MB**
S87	0.00	1.1831	1.1843	16.01MB
F88	**0.00 (EMBED=254)**	**1.1820**	**1.1839**	**16.00MB — FITS**

WD=0 optimal at 10L 4× — opposite to 26L result. Wider MLP needs full weight freedom.

Binary Quantisation Track

Motivation

Binary quantisation constrains weights to {-1, +1} with no zero state. At 1 bit/param vs ternary's 1.6 bits/param, binary packs approximately 60% more parameters per MB. The hypothesis was that additional depth could compensate for the loss of the zero state.

Starting point: the ternary best config (10L, 768d, 8h, 4kv, 4× relu², FP8, 524k batch, 599s) scoring 1.1578 sliding bpb.

Binary Scaling Runs

Run	Layers	MLP	FP	Other	Steps	ms/step	Sliding bpb	Artifact	Fits
F17	17	4×	FP8	—	4010	149	1.2022	17.45MB	No
F1	**14**	**4×**	**FP8**	**—**	**4820**	**124**	**1.1824**	**14.74MB**	**Yes**
F2	14	4×	FP8	EMA	4800	125	1.2110	14.56MB	Yes
S3	15	4×	FP8	—	1000	133	1.3114	15.65MB	Yes
S4	20	3×	FP8	—	1000	160	1.3077	16.90MB	No
S5	21	3×	FP4	—	1000	167	1.3676	16.64MB	No
S6	19	3×	FP8	—	1000	152	1.3130	16.16MB	No
S7	15	4×	FP8	refiner	1000	135	1.3123	15.89MB	Yes
S8	15	4×	FP8	smear	1000	155	1.3043	15.67MB	Yes
S9	15	4×	FP8	tversky_attn	1000	179	1.4016	15.74MB	Yes

Key Decisions from Binary Scaling

MLP width (4× vs 3×): 4× won even when 3× received 4–5 extra layers. S3 (15L 4×) outperformed S6 (19L 3×) at matched steps. Width matters more than depth past a minimum viable layer count.

FP storage (FP8 vs FP4): FP4 added a 0.06 bpb roundtrip penalty and was immediately ruled out. FP8 used for all non-binary tensors.

Layer count: 17L was the theoretical maximum at 4× FP8 but landed 1.45MB over budget. 15L at 15.65MB was the maximum that fit. 14L left 1.26MB headroom.

EMA: Mathematically sound for binary (no zero bucket means `mean(|Q|)=1.0` always, clean roundtrip). In practice, 0.03 bpb worse — the smoothed weights apparently hurt binary's learning dynamics despite the clean quantisation math.

Smear: 0.007 bpb gain at 1000 steps but added 22ms/step overhead (133→155ms). Retained for the extended binary run to test whether the gain survives the step penalty at longer training.

Refiner (causal conv): Neutral at 1000 steps, added 2ms/step. Not justified.

Tversky attention projection: 0.09 bpb worse. Completely incompatible with binary weights.

Activation: relu² inherited from ternary sweeps, not retested for binary. SwiGLU would cost ~4MB extra across 15 layers, eliminating the layer budget advantage.

Extended Binary Run (Unconstrained Compute)

To measure the binary architecture's convergence ceiling without the 10-minute wallclock constraint, a single extended run was conducted at 50,000 steps (~2 hours on 8×H100).

Configuration: 15L 768d, 4× relu², FP8, smear, 524k batch tokens, seed=42, MUON_WD=0.0

```
step:50000/50000 val_loss:2.9692 val_bpb:1.1497 train_time:7763s
artifact:15.60MB binary:97320960(13685760B) fp:2542200(2585072B) code:70399
budget:15670651/16000000 (15.67/16.00MB) FITS
final_binary_roundtrip val_loss:2.9743 val_bpb:1.1516
temp_scaling optimal_T:0.90
final_sliding val_loss:2.9027 val_bpb:1.1239 (stride=16, T=0.90)
```

Metric	Value
val_bpb	1.1497
RT bpb	1.1516
Sliding bpb	**1.1239**
Artifact	15.60MB (15.67MB total)
Params	97,320,960
Steps	50,000
ms/step	155.3
Training time	~2.15 hours

The 1.1239 sliding bpb demonstrates that with sufficient compute the binary architecture reaches strong quality. This validates the compression approach — nearly 100M parameters in 15.67MB via 1-bit quantisation — though the 50k steps required far exceeds the competition's 10-minute budget.

Binary vs Ternary at Equal Architecture (Dev Scale)

Metric	Binary	Ternary	Delta
val_bpb	1.8609	1.8113	Ternary wins by 0.050
Artifact	9.14MB	11.56MB	Binary saves 2.42MB
ms/step	918	924	Identical
RT gap	0.000	0.000	Both clean

Ternary is better at equal architecture. Binary's only advantage is fitting more layers in the same budget.

Binary Conclusion

Binary lost the depth-for-sparsity trade. The 5 extra layers (15L binary vs 10L ternary) could not overcome ternary's representational advantage from the zero state. The 0.0016 bpb gap measured at 500 dev steps significantly understated the true difference at convergence. Ternary at 1.1578 sliding bpb (10-minute budget) outperforms

binary's best fitting run (F1: 1.1824 at 14L without smear) by 0.025 bpb. Even the over-budget 17L binary run (1.2022) could not match ternary.

The extended 50k-step binary run reaching 1.1239 sliding bpb shows that binary has a competitive convergence ceiling, but it requires approximately 8× more training steps to approach competitive quality — well beyond the competition constraints.

Grouped MLP Investigation

Tested GroupedTernaryLinear: splits MLP into independent groups for parameter/speed savings.

Real Model Results (relu² 3×, 768d, 600s)

Run	Config	Layers	ms/step	Steps	val_bpb	Artifact
F64	standard	12	99.4	6030	1.1873	17.50MB
F72	g=2	12	87.4	6870	1.2180	12.97MB
F71	g=4	12	83.5	7190	1.2429	10.74MB
F73	g=2	16	114.2	5260	1.2037	16.04MB
F74	swiglu g=2	12	113.3	5300	1.2084	15.24MB

Cross-group isolation costs 0.031–0.056 bpb. Even with 4 extra layers (F73), only recovers 0.014 of the deficit. **Not viable for language modelling.**

Differential Attention

Microsoft (2024): computes two attention maps from split Q/K and takes their difference.

Run	Config	ms/step	Steps	val_bpb
F64	standard	99.4	6030	1.1873
F68	diff_attn	109.3	5480	1.2094

Splits 96-dim heads into 48-dim sub-heads — insufficient dimensionality for meaningful attention patterns at this model scale.

Sequence Refiner (CausalConvRefiner)

Run	Config	ms/step	Steps	val_bpb	Artifact
F64	none	99.4	6030	1.1873	17.50MB
F69	k=3	102.2	5860	1.1885	19.92MB
F70	k=5	103.0	5820	1.2018	18.13MB

Noise-level quality improvement with storage bloat. 12 attention layers already saturate local pattern capture.

ByteCNN Vocabulary Generator

Replaces `nn.Embedding(8192, 256)` with a CNN that generates the embedding matrix from byte spellings.

```
step:500 loss:9.0471 — step:2000 loss:9.0471 (flat, no learning)
```

All 8192 CNN-generated embeddings converge to near-identical vectors at initialisation. The CNN's inductive bias (byte-similar tokens → similar embeddings) destroys the initial diversity needed for gradient signal.

Asymmetric Tokenizer Investigation

8k BPE input with 256-byte output to eliminate large output projection.

Model	BPB	Notes
Standard (tied, emb=256)	3.10	reference
Asymmetric parallel (emb=256)	8.65	byte independence assumption fails
Asymmetric autoregressive (emb=256)	8.17	tiny GRU insufficient capacity

Multi-byte parallel heads assume conditional independence between bytes within a token — mathematically incorrect. Sequence-length mismatch (7 BPE tokens → 70 bytes) also incompatible with the evaluation framework.

Linear Alternative Exploration

Systematic notebook testing of linear layer alternatives at real model dimensions (768d).

Projection Benchmark (DIM → DIM, H100)

Model	Params	ms	vs Linear
Linear	589,824	0.07ms	1.00×
LowRank r=64	98,304	0.03ms	0.44×
BlockDiag b=4	147,456	0.03ms	0.40×
Grouped g=4	147,456	0.03ms	0.40×
BD4 + mix32	196,608	0.07ms	0.97×
Hash 65536	65,536	0.08ms	1.13×

BlockDiag/Grouped offer speed advantages but cross-group isolation degrades LM quality in practice.

H100 Microbenchmark Results

Standalone kernel timing vs torch.compile behaviour (critical lesson: standalone microbenchmarks can mislead when torch.compile fuses operations).

STE Speed

Variant	ms/call
Current	0.041

Variant	ms/call
Reciprocal	0.043

No gain — 48 STE calls/step = ~2ms overhead (unavoidable).

Contiguous Checks

Q and K are contiguous after RoPE. V is non-contiguous (view into fused QKV). V's `.contiguous()` costs 0.065ms/call = 0.78ms/step (necessary for flash_attn).

RoPE Variants

Current (half-split + cat) is fastest at 0.52ms/call.

Softcap: Poly5 vs Tanh

Variant	ms/call
Poly5 (current)	8.43
Poly3	5.98
Tanh	2.12
Hardtanh	0.71

Critical finding: Tanh is 4× faster standalone due to H100 hardware transcendental units. However in the real training loop, torch.compile fuses poly5 with surrounding ops into a single kernel. **Switching to tanh broke fusion — F63 was 16ms/step slower.** Poly5 retained.

CE + Z-Loss Fusion

Variant	ms/call (fwd+bwd)
Separate (current)	16.56
Fused (shared LSE)	12.33

Same lesson: 4.2ms saving standalone, but torch.compile already optimises `F.cross_entropy`. Manual gather+logsumexp prevents optimisation. Current approach retained.

Efficiency Analysis

BPB Gained Per Component

Component	BPB gain	Source
relu → relu²	−0.024	F55 vs F56
MLP 2× → 3× (relu²)	−0.017	F56 vs F64
MLP 3× → 4× (relu²)	−0.008	F64 vs F75
relu² → swiglu (at 3×)	−0.010	F64 vs F59
+1 layer (average)	−0.0012	scaling data
fp16 → fp8 (RT penalty)	+0.002	run 42 vs 49
Sliding eval stride=16	−0.025	F22 data
WD=0.04 vs WD=0 (at 26L)	−0.001	F7 vs F6

MB Cost Per Component

Component	MB/layer
relu² 2× layer	0.767
relu² 3× layer	1.003
relu² 4× layer	1.220
swiglu 3× layer	1.357
fp16 → fp8 (fixed saving)	–2.51

Efficiency Ratio (BPB Gained Per MB Spent)

Change	BPB gain	MB cost	BPB/MB
relu → relu²	–0.024	0.00	infinite (free)
Sliding eval	–0.025	0.00	infinite (free)
MLP 2× → 3×	–0.017	+2.83 (12L)	–0.0060/MB
MLP 3× → 4×	–0.008	+2.83 (12L)	–0.0028/MB
relu² → swiglu	–0.010	+4.25 (12L)	–0.0024/MB
+1 layer (relu² 2×)	–0.0012	+0.767	–0.0016/MB
+1 layer (relu² 3×)	–0.0012	+1.003	–0.0012/MB

MLP 2×→3× is the most efficient paid upgrade. relu² and sliding eval are free wins.

Layer Budget at 768d

Config	Max Layers	Est ms/step
relu² 2× fp16	14L	~95ms
relu² 2× fp8	17L	~97ms
relu² 3× fp16	10L	~99ms
relu² 3× fp8	13L	~106ms
relu² 4× fp8	10L	~92ms
swiglu 3× fp8	9L	~105ms

Ternary-Incompatible Techniques

These are not merely unhelpful but structurally incompatible with 1.58-bit quantisation:

Technique	Mechanism of failure
EMA	**Weight averaging → values cluster near zero → ternary rounds most to 0 → 0.12 bpb RT gap**
TTT-LoRA	**LoRA delta computed outside RMSNorm space that TernaryLinear normalises into. Corrupts calibrated representations at convergence**
Ternary prototypes + sigmoid	**Sigmoid membership needs continuous values. Ternary {-1,0,+1} collapses membership patterns → 0.077 RT gap**

Technique	Mechanism of failure
LM head rank factorisation	**SVD factors U,V need fp16 precision. Storage exceeds original tied embedding**

Software Optimisations

Optimisation	Saving	Notes
Fused QKV (c_q+c_k+c_v → single matmul)	~2ms/step	Safe: in_features divisible by all group sizes
Fused SwiGLU/relu² (gate+up → single wide matmul)	~2-4ms/step	Same params, fewer kernel launches
Z-loss regularisation (1e-4 x logsumexp²)	quality	Anchors logits, keeps STE gradients sharp
DataLoader int16 transfer (pin then cast on GPU)	~1ms/step	4× less PCIe bandwidth
FlashAttention-3	~13ms/step	~9% speedup, ~380 free training steps
TernaryLinear bf16 weights, cleaner STE	~1ms/step	Eliminates fp32 roundtrip
DDP static_graph + gradient_as_bucket_view	~1ms/step	Free when find_unused=False
Fused optimizer loop (LR set + step in one pass)	~0.5ms/step	Fewer Python-level iterations
Removed CUBLAS determinism tax	~1ms/step	Not required for competition
Temperature grid: 5 points instead of 21	~1s total	T=0.90 consistently with relu²
Temp scaling moved to eval phase	~3 steps gained	No longer steals training time
`_e()` helper for Hyperparameters	-1.8KB code	Eliminates env var boilerplate
3D tensor ternary quantisation	storage fix	Conv1d weights reshaped to 2D for ternary

Rejected Techniques (Summary)

Technique	Reason
Tversky (all variants)	Quality-neutral on FineWeb LM — confirmed via synthetic data analysis; speed penalty with relu²
Differential attention	Halved head_dim (96→48) degrades quality at this model scale
Grouped MLP (g=2, g=4)	Cross-group isolation costs 0.031–0.056 bpb; not recoverable with extra layers
CausalConvRefiner	Noise-level quality; storage bloat from Conv1d weights
ByteCNN vocabulary generator	Embedding collapse — CNN inductive bias destroys initial diversity
Asymmetric tokenizer	Byte independence assumption incorrect; sequence mismatch with eval framework
EMA	Incompatible with ternary — weight averaging causes 0.12 bpb RT gap
TTT-LoRA	Architectural incompatibility with RMSNorm space in TernaryLinear
LM head factorisation	SVD factors bloat artifact beyond budget; unrecoverable quality loss
MTP	0.006 bpb worse — model capacity too limited for auxiliary objectives

Technique	Reason
BigramHash	0.020 bpb worse at convergence; fp16 table displaces ternary layers
Seq/batch schedule	Recompile and step penalties dominate at 600s wallclock
SmearModule	+22% step cost for −0.001 gain within ternary 10-minute budget
Depth recurrence	Halves effective steps; OOM at DR=3
AdamW for matrix params	Clearly inferior to Muon for ternary weights
FP4 storage	0.026–0.029 RT gap even with QAT — unrecoverable
Tanh softcap	Faster standalone but breaks torch.compile kernel fusion
Fused CE+Z-loss	Same — breaks compile optimisation
16 heads at 768d	48-dim head_dim insufficient for meaningful attention
relu (plain)	Strictly dominated by relu²
leaky relu	Strictly dominated by relu²
Distillation (in-run)	Train-from-scratch teacher always worse than supervised
reduce-overhead compile	Rotary + embed_proj_rev incompatible with CUDA graphs
max-autotune compile	30+ minute kernel search prohibitive for 600s runs
Skip weights zero-init	0.010 bpb worse — decoder needs skip signal from step 0
EMBED_DIM=0 (full 512)	19.78MB artifact — 3.78MB over budget
Untie lm_head full-rank	7.3MB budget overrun not justified by 0.005 bpb gain

Decision Log

Decision	Rationale
8k vocabulary	−0.42 bpb, largest single win
relu² activation	−0.024 bpb vs relu, free (no cost)
4×MLP width	Best BPB within budget at 10L; 0.008 better than 3×
10L 768d	Minimum viable depth at 768d with maximum MLP width
WD=0.0 at 10L 4×	Opposite to deep models — wider MLP needs full weight freedom
fp8 storage	Halves fp_params (5MB→2.5MB), enables wider MLP within budget
EMBED_DIM=254	256-2 dims to fit artifact+code under 16,000,000 byte budget; ~0.0004 bpb cost
BITNET_GROUP_SIZE=128	Same quality as 64; saves 0.69MB
8 heads, 4 KV, 96-dim head_dim	16h at 48-dim insufficient; MHA only +0.0012 at +1.5MB
Poly softcap	Fuses with torch.compile; tanh breaks fusion
ROPE_BASE=5000 + YaRN 2048	Best frequency calibration
Muon optimizer	Newton-Schulz normalisation compensates for ternary STE gradient attenuation
MUON_BACKEND_STEPS=3	Equivalent to 5 at convergence; +190 extra steps
MUON_MOMENTUM=0.95	Both directions degrade; affects artifact via zero_frac
WARMDOWN=20%	Asymmetric — too little hurts more than too much
MATRIX_LR=0.04	Higher LR compensates for ternary STE gradient attenuation
SCALAR_LR=0.02	Optimal — scalars do not pass through STE
TIED_EMBED_LR=0.02	Optimal

Decision	Rationale
TRAIN_BATCH_TOKENS=524k	Optimal tradeoff between gradient quality and step count
Base-3 + LZMA	39% reduction over int8+zlib
Shrinkage fix	Eliminates all RT gaps universally
Skip weights ones-init	Decoder needs skip signal from step 0; zeros costs 0.010 bpb
Tied embeddings	Untie costs 7.3MB; not justified
Standard attn projection	Tversky quality-neutral; grouped destroys quality
No EMA	Fundamentally incompatible with ternary
No TTT	RMSNorm space incompatibility confirmed across 6 runs
No MTP	Confirmed post-fix: 0.006 bpb worse
Temperature scaling T=0.90	relu² logits slightly underconfident; auto-calibrated
Fused QKV + relu²	~130-180 free training steps per run
Z-loss regularisation	Anchors logits; keeps STE gradients sharp
FlashAttention-3	Free ~380 extra training steps per 600s run
Sliding eval stride=16	Best quality when eval budget unconstrained
Optimizer coverage fix	embed_proj/embed_proj_rev now train; +0.055 bpb improvement
MAX_WALLCLOCK_SECONDS=599	1s leeway for safety margin
Binary 15L 768d 4× fp8	97M params in 15.67MB — maximum parameter density; convergence ceiling validated at 50k steps